IIT JAM MATHEMATICS – LINEAR ALGEBRA CHEAT SHEET

Matrices · Finite Dimensional Vector Spaces · Linear Transformations · Systems of Equations · Eigenvalues & Diagonalizability

1. Vector Spaces & Subspaces

- **Subspace Test:** $W \subseteq V$ is a subspace $\iff \alpha u + v \in W$ for all $\alpha \in \mathbb{F}, u, v \in W$ (and $W \neq \emptyset$).
- **Intersection & Sum:** If $W_1, W_2 \leq V$:
  - $W_1 \cap W_2$ is always a subspace.
  - $W_1 \cup W_2$ is a subspace $\iff W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.
  - $W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$ is the smallest subspace containing $W_1 \cup W_2$.
- **Dimension Theorem for Subspaces:**

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

- **Direct Sum ($V = W_1 \oplus W_2$):**
  - $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$.
  - Every $v \in V$ can be *uniquely* written as $w_1 + w_2$.
  - $\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2)$.
- **Standard Dimensions over $\mathbb{R}$:**
  - $\dim(M_{m \times n}(\mathbb{R})) = mn$
  - Symmetric $n \times n$: $\frac{n(n+1)}{2}$
  - Skew-Symmetric $n \times n$: $\frac{n(n-1)}{2}$
  - Trace zero $n \times n$: $n^2 - 1$
  - $\mathcal{P}_n(\mathbb{R})$ (poly of degree $\leq n$): $n + 1$
  - $\mathbb{C}^n$ over $\mathbb{R}$: $2n$; $\mathbb{C}^n$ over $\mathbb{C}$: n.

2. Linear Independence

- **Linear Independence (LI):** $\{v_1, \dots, v_k\}$ is LI if $\sum_{i=1}^{k} c_i v_i = 0 \implies c_1 = \cdots = c_k = 0$.
- **Linear Dependence (LD):** At least one vector is a linear combination of the preceding vectors. Any set containing the zero vector $\mathbf{0}$ is LD.
- **Basis:** A linearly independent spanning set of V.
  - Minimal generating set $\equiv$ Maximal LI set.
  - Every vector $v \in V$ is uniquely expressible as a linear combination of basis elements.
- **Key Dimension Properties (for $\dim(V) = n$):**
  - Any subset with $> n$ vectors is LD.
  - Any LI subset with n vectors is a basis.
  - Any spanning set with n vectors is a basis.
  - Every LI set can be extended to a basis.
  - If $W \leq V$, then $\dim(W) \leq \dim(V)$; $\dim(W) = \dim(V) \iff W = V$.
- **Wronskian Test for Functions:** If $W(f_1, \dots, f_n)(x_0) \neq 0$ for some x_0, the functions are LI.

3. Linear Transformations (LT)

- **Definition:** $T : V \to W$ satisfies $T(\alpha u + v) = \alpha T(u) + T(v)$.
- **Fundamental Spaces:**
  - $\text{Null}(T) = \ker(T) = \{v \in V : T(v) = 0\} \leq V$
  - $\text{Range}(T) = \text{Im}(T) = \{T(v) : v \in V\} \leq W$
  - $\text{nullity}(T) = \dim(\text{Null}(T))$, $\text{rank}(T) = \dim(\text{Range}(T))$
- **Rank-Nullity Theorem (Dimension Theorem):**

$$\boxed{\text{rank}(T) + \text{nullity}(T) = \dim(V)}$$

- **Injective & Surjective Criteria:**
  - T is one-to-one (injective) $\iff \text{Null}(T) = \{0\} \iff \text{nullity}(T) = 0$.
  - T is onto (surjective) $\iff \text{Range}(T) = W \iff \text{rank}(T) = \dim(W)$.
  - If $\dim(V) = \dim(W) < \infty$:

    T is injective $\iff$ T is surjective $\iff$ T is bijective (inv

  - If $\dim(V) < \dim(W) \implies T$ cannot be surjective.
  - If $\dim(V) > \dim(W) \implies T$ cannot be injective.
- **Image of LI sets:**
  - S is LI and T is injective $\implies T(S)$ is LI.
  - $V = \text{span}(S) \implies \text{Range}(T) = \text{span}(T(S))$.

4. Matrix Representation & Change of Basis

- **Matrix of T:** Let $\beta = \{v_1, \dots, v_n\}$ for V and $\gamma = \{w_1, \dots, w_m\}$ for W.

$$[T]_\beta^\gamma = \big[[T(v_1)]_\gamma \quad [T(v_2)]_\gamma \quad \cdots \quad [T(v_n)]_\gamma\big]_{m\times n}$$

- **Evaluation Property:** $[T(x)]_\gamma = [T]_\beta^\gamma [x]_\beta$.
- **Composition:** $[U \circ T]_\alpha^\gamma = [U]_\beta^\gamma [T]_\alpha^\beta$.
- **Change of Basis:** For $T : V \to V$ and bases β, β':

$$[T]_{\beta'} = P^{-1}[T]_\beta P \quad \text{where } P = [I]_{\beta'}^\beta$$

- **Similar Matrices ($A \sim B$):** $B = P^{-1}AP$ for invertible P.
- **Similarity Invariants:** If $A \sim B$, they share:
  - $\det(A) = \det(B)$ and $\operatorname{trace}(A) = \operatorname{trace}(B)$.
  - $\operatorname{rank}(A) = \operatorname{rank}(B)$ and $\operatorname{nullity}(A) = \operatorname{nullity}(B)$.
  - Same characteristic & minimal polynomials.
  - Same eigenvalues with identical AM and GM.

5. Systems of Linear Equations (Ax

Let $A \in M_{m\times n}(\mathbb{F})$, $x \in \mathbb{F}^n$, $b \in \mathbb{F}^m$, and augmented matrix $\tilde{A} = [A \mid b]$.

- **Rouché–Capelli Theorem (Consistency):**
  - **Consistent** $\iff \operatorname{rank}(A) = \operatorname{rank}([A \mid b])$.
  - **Inconsistent** $\iff \operatorname{rank}(A) < \operatorname{rank}([A \mid b])$.
- **Structure of Solution Set:**

Condition	Rank Relation	Solut
$\operatorname{rank}(A) \neq \operatorname{rank}(\tilde{A})$	$\operatorname{rank}(A) < \operatorname{rank}(\tilde{A})$	No so
$\operatorname{rank}(A) = \operatorname{rank}(\tilde{A}) = n$	Full column rank	Uniqu
$\operatorname{rank}(A) = \operatorname{rank}(\tilde{A}) = r < n$	$n - r$ free params	∞ sol

- **General Solution Form:** $x = x_p + x_h$, where $Ax_p = b$ (particular) and $x_h \in \operatorname{Null}(A)$ (homogeneous).
- **Homogeneous System ($Ax = 0$):**
  - Dimension of solution space $= \operatorname{nullity}(A) = n - \operatorname{rank}(A)$.
  - Non-trivial (non-zero) solution exists $\iff \operatorname{rank}(A) < n$.
  - For $n \times n$ system: Non-trivial solution $\iff \det(A) = 0$.

6. Matrix Rank & Inverses

- $\operatorname{rank}(A)$ = Row Rank = Column Rank = Number of non-zero rows in RREF.
- **Rank Inequalities:**
  - $\operatorname{rank}(A_{m\times n}) \le \min(m, n)$.
  - $\operatorname{rank}(A + B) \le \operatorname{rank}(A) + \operatorname{rank}(B)$.
  - $\operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B))$.
  - **Sylvester's Inequality:**

$$\operatorname{rank}(A) + \operatorname{rank}(B) - n \le \operatorname{rank}(AB) \quad (\text{for } A_{m\times n}, B_{n\times p})$$

  - $\operatorname{rank}(A^TA) = \operatorname{rank}(AA^T) = \operatorname{rank}(A)$ (over $\mathbb{R}$).
- **Invertibility ($n \times n$ Matrix A):** A^{-1} exists $\iff \det(A) \neq 0 \iff \operatorname{rank}(A) = n \iff \operatorname{Null}(A) = \{0\} \iff 0$ is not an eigenvalue of A.
- **Formula for Inverse:** $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$.
- **Rank of Adjugate:**

$$\operatorname{rank}(\operatorname{adj}(A)) = \begin{cases} n, & \text{if } \operatorname{rank}(A) = n \\ 1, & \text{if } \operatorname{rank}(A) = n - 1 \\ 0, & \text{if } \operatorname{rank}(A) \le n - 2 \end{cases}$$

- $\operatorname{adj}(AB) = \operatorname{adj}(B)\operatorname{adj}(A)$; $\det(\operatorname{adj}(A)) = (\det(A))^{n-1}$.
- $\operatorname{adj}(\operatorname{adj}(A)) = (\det(A))^{n-2}A$.

7. Determinant Shortcuts & Rules

- $\det(cA) = c^n \det(A)$ for $A \in M_{n\times n}(\mathbb{F})$.
- $\det(AB) = \det(A)\det(B) = \det(BA)$.
- $\det(A^{-1}) = \frac{1}{\det(A)}$; $\det(A^T) = \det(A)$.
- **Block Matrices:** If A, D are square:

$$\det\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det(A)\det(D)$$

$$\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A)\det(D - CA^{-1}B) \quad (\text{if } A^{-1} \text{ exists})$$

- **Vandermonde Determinant:**

$$\det\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$$

- If $A = uv^T$ (rank 1 matrix), $\det(I_n + uv^T) = 1 + v^Tu = 1 + \operatorname{trace}(A)$.
- $\det(A) = \prod_{i=1}^n \lambda_i, \quad \operatorname{trace}(A) = \sum_{i=1}^n \lambda_i$.

8. Eigenvalues & Eigenvectors

- **Characteristic Equation:** $\chi_A(\lambda) = \det(A - \lambda I) = 0$.
- **For 2×2 Matrix:** $\lambda^2 - \operatorname{trace}(A)\lambda + \det(A) = 0$.
- **For 3×3 Matrix:**

$$\lambda^3 - \operatorname{trace}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - \det(A) = 0$$

(where M_{ii} are the principal 2×2 minors).
- **Eigenspace E_λ:** $E_\lambda = \operatorname{Null}(A - \lambda I)$.
- **Multiplicities:**
  - $\operatorname{AM}(\lambda)$: Multiplicity of λ as a root of $\chi_A(\lambda)$.
  - $\operatorname{GM}(\lambda) = \dim(E_\lambda) = n - \operatorname{rank}(A - \lambda I)$.
  - Fundamental Inequality: $1 \le \operatorname{GM}(\lambda) \le \operatorname{AM}(\lambda)$.
- **Eigenvalue Transformations:**

Matrix	Eigenvalues
A^k	λ^k
A^{-1}	λ^{-1}
$f(A)$ (polynomial)	$f(\lambda)$
$\operatorname{adj}(A)$	$\frac{\det(A)}{\lambda}$ $(\lambda \neq 0)$
$cA + dI$	$c\lambda + d$

- Distinct eigenvalues have *linearly independent* eigenvectors.

9. Cayley-Hamilton & Minimal Polynomial

- **Cayley-Hamilton Theorem:** Every square matrix satisfies its own characteristic equation: $\chi_A(A) = 0$.
- **Computation of A^{-1} and A^k:** If $\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0 = 0$:

$$A^{-1} = -\frac{1}{c_0}\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I\right)$$

- **Minimal Polynomial $m_A(x)$:**
  - Unique monic polynomial of lowest degree such that $m_A(A) = 0$.
  - $m_A(x)$ divides $\chi_A(x)$.
  - $\chi_A(x)$ and $m_A(x)$ share the *exact same distinct roots.*

10. Diagonalizability

- $A_{n\times n}$ is diagonalizable $\iff A = PDP^{-1}$ where $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$ and columns of P are n LI eigenvectors.
- **Necessary & Sufficient Conditions:**
  1. $\text{GM}(\lambda_i) = \text{AM}(\lambda_i)$ for every eigenvalue λ_i.
  2. The minimal polynomial $m_A(x)$ splits into **distinct linear factors**:

$$m_A(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_k)$$

- **Sufficient Condition:** If A has n *distinct* eigenvalues, it is always diagonalizable. (Converse is false!)
- **Simultaneous Diagonalizability:** A, B are simultaneously diagonalizable $\iff AB = BA$ (they commute) and both are diagonalizable.

11. Special Classes of Matrices

- **Idempotent ($A^2 = A$):**
  - Eigenvalues $\in \{0, 1\}$.
  - $\text{trace}(A) = \text{rank}(A)$.
  - Always diagonalizable: $m_A(x) \mid x(x-1)$.
  - $I - A$ is also idempotent; $\text{Null}(A) = \text{Range}(I - A)$.
- **Nilpotent ($A^k = 0$ for some k):**
  - All eigenvalues $= 0$.
  - $\text{trace}(A) = 0$ and $\det(A) = 0$.
  - Characteristic polynomial: $\chi_A(\lambda) = \lambda^n \implies A^n = 0$.
  - Diagonalizable $\iff A = 0$.
  - $I + A$ is always invertible.
- **Involutory ($A^2 = I$):**
  - Eigenvalues $\in \{1, -1\}$.
  - Always diagonalizable: $m_A(x) \mid (x-1)(x+1)$.
- **Real Symmetric ($A^T = A$):**
  - All eigenvalues are **purely real**.
  - Eigenvectors of distinct eigenvalues are orthogonal.
  - Always **orthogonally diagonalizable**: $A = PDP^T$ ($P^TP = I$).
- **Real Skew-Symmetric ($A^T = -A$):**
  - Eigenvalues are 0 or **purely imaginary** ($\pm ib$).
  - If n is odd $\implies \det(A) = 0$ and $\text{rank}(A)$ is even.
- **Orthogonal ($A^TA = I$):**
  - $|\det(A)| = \pm 1$.
  - Eigenvalues satisfy $|\lambda| = 1$ ($\lambda = e^{i\theta}$).

12. High-Yield IIT JAM Exam Shortcuts

- **All-Ones Matrix (J_n):** $\text{rank}(J_n) = 1$. Eigenvalues are n (mult 1) and 0 (mult $n-1$). $\chi_A(\lambda) = \lambda^{n-1}(\lambda - n)$.
- **Constant Row Sum s:** If every row sum of A equals s, then:
  - $\lambda = s$ is an eigenvalue with eigenvector $\begin{bmatrix}1 & 1 & \cdots & 1\end{bmatrix}^T$.
  - Row sum of A^k is s^k.
- **Circulant Matrices:** $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \implies \lambda = a+b, a-b$.
- **Rank 1 Matrix ($A = xy^T$):** $\text{trace}(A) = x^Ty = y^Tx$. Eigenvalues are $\text{trace}(A)$ (mult 1) and 0 (mult $n-1$). Diagonalizable $\iff \text{trace}(A) \neq 0$.
- **Commutator $[A, B] = AB - BA$:** $\text{trace}(AB - BA) = 0 \implies AB - BA \neq I_n$.
- **Companion Matrix of $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$:** $\chi_A(x) = m_A(x) = p(x) \implies \text{GM}(\lambda) = 1$ for all λ.